

Magnitude, Not Shape: A Survivorship-Free, Significance-Tested Evaluation of Downside-Risk Metrics

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ABSTRACT

Every few years a new downside- or tail-risk metric is proposed to improve on plain return volatility, always on the same premise: the *shape* of the loss distribution — its asymmetry, tail heaviness, or co-crash structure — carries information that symmetric volatility misses. We test that premise as adversarially as the data allow. On a survivorship-free (a crypto panel retaining delisted coins, plus equity), multi-regime, multiple-testing-corrected evaluation of ten metrics across five test beds, no metric — published or machine-discovered — reliably beats volatility at forecasting an asset’s forward maximum drawdown. The anatomy of that failure is a clean law: *magnitude, not shape*. The partial rank correlation of Value-at-Risk with forward drawdown, controlling for volatility, survives Benjamini–Hochberg correction on all five beds (up to 0.19), while every tail-*shape* metric adds nothing or subtracts. We explain why volatility is hard to beat (it is persistent, though a predictive mediation test shows not a sufficient statistic), show the tiny magnitude edge is not economically free (a cost-aware backtest and a pre-registered composite add nothing over volatility except in fat-tailed crypto), and demonstrate that an autonomous large-language-model discovery loop only inflates validation while its locked-test score plateaus at volatility — a data-snooping caution as agentic research makes signal search nearly free. Volatility is not glamorous, but on a level field it is very hard to beat.

CCS CONCEPTS

• Computing methodologies → Machine learning; • Applied computing → Economics.

KEYWORDS

downside risk, tail risk, drawdown, volatility, survivorship bias, multiple testing, data snooping, LLM agents

1 INTRODUCTION

The market’s dangers are asymmetric: prices fall faster than they rise, and it is the fall that ruins investors. From this intuition a large literature has grown, proposing metrics that look past symmetric volatility to the *shape* of the loss distribution — lower partial moments and the Sortino ratio [6, 13, 19], the partial-moment “predictive” ratio we originally set out to audit [20], left-tail Value-at-Risk and expected shortfall [3, 18], realized semibetas [8], downside beta [2, 16], lower-tail dependence [9], disappointment-aversion risk [12], and tail-exponent factors [15]. Each rests on the same premise: that asymmetry or tail structure carries information volatility misses.

We test that premise on the fairest field we can build. Three disciplines the source literature usually omits turn out to matter

enormously: a *survivorship-free* universe (so the assets that actually died are present), a *locked* out-of-sample holdout (so search cannot leak into the verdict), and *multiple-testing correction* (so an evaluation of dozens of comparisons does not manufacture winners). We evaluate ten metrics as forecasters of 90-day forward maximum drawdown across five cross-sectional test beds spanning calm, bull, and crash regimes in crypto and equity, with per-date rank-correlation skill, paired block-bootstrap confidence intervals, and Benjamini–Hochberg (BH) correction.

The headline is negative and, in hindsight, unsurprising: no metric reliably beats plain volatility. Our contribution is the *anatomy* of that result and the honesty of the battery behind it. (1) A clean empirical law — *magnitude, not shape*: the only signal surviving correction beyond volatility is the *size* of tail losses (VaR/ES/downside deviation), not their asymmetry (§4). (2) Volatility is hard to beat because it is *persistent*, though a predictive test shows forward volatility is not a sufficient statistic (§5). (3) The residual edge is not economically free, except in fat-tailed crypto (§5). (4) An autonomous large-language-model (LLM) discovery loop cannot beat volatility out-of-sample and only overfits validation — a data-snooping caution as agentic search makes signal-hunting nearly free (§5). Every experiment is retained; supporting detail is in the appendices.

Related work. Lower partial moments generalize semi-variance [6, 13] and underpin the Sortino ratio [19] and the partial-moment ratio we audit [20]; tail measures include VaR/ES [3, 18], semibetas [8], downside beta [2, 16], lower-tail dependence [9], disappointment aversion [12], and tail exponents [15]. Most were built to *price* the cross-section, not forecast one asset’s drawdown; a separate literature studies the drawdown statistic directly [10, 17]. That volatility is persistent is the founding fact of ARCH/GARCH [1, 7, 11]; that searching many signals inflates in-sample skill is classical data snooping [4, 5, 14, 21]. Our novelty is to show an *autonomous* agent reaches the snooping frontier at near-zero cost.

2 DATA AND PROTOCOL

Five test beds. *Crypto (survivorship-free)*: a CoinMarketCap daily panel (2013–2018) that retains crashed and delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes — 2016 (calm), 2017 (bull), 2018 (crash). *Equity*: S&P 500 constituents, 1994–1999 (calm) and 2005–2009 (GFC). The equity beds are survivorship-biased by construction — freely available equity data retains almost no true deaths, itself a telling finding — so the survivorship claim rests on the crypto beds, while equity tests cross-asset generality.

Skill and significance. From a 180-day trailing window we forecast the 90-day forward maximum drawdown of each

Table 1: Standalone out-of-sample drawdown-forecast Spearman correlation. * = significantly beats volatility, † = significantly worse (both survive BH, $q = 0.05$); unmarked = indistinguishable. The Hill tail exponent [15] is omitted here (not estimable per date on equity); its increment appears in Table 2.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
volatility (base)	0.62	0.36	0.38	0.61	0.56
downside deviation	0.62	0.38*	0.41	0.58†	0.54†
VaR	0.61	0.40*	0.36	0.58†	0.54†
ES [3]	0.62	0.35	0.33†	0.58†	0.52†
neg. semibeta [8]	0.40†	0.26†	0.31†	0.49†	0.46†
downside beta [2]	0.01†	0.09†	0.15†	0.24†	0.34†
lower-tail dep. [9]	-0.38†	-0.15†	-0.29†	-0.07†	-0.05†
GDA [12]	-0.06†	-0.11†	-0.05†	-0.24†	-0.32†
UPM/LPM [20]	-0.55†	-0.28†	-0.35†	-0.43†	-0.42†

asset (delisting scored as a -100% loss); skill is the per-date cross-sectional Spearman correlation between metric and drawdown, averaged over the bed, with paired block-bootstrap 95% CIs — cross-sectional (not time-series) because the operational question is which assets to derisk on a date. The partial Spearman controlling for volatility isolates what a metric adds *beyond* it. Significance flags are BH-corrected ($q = 0.05$) across the ~45 metric $\times$ bed cells; the discovery experiment (§5) uses a strict train/validation/*locked-test* split scored once after the metric is frozen. The verdict, the magnitude-vs-shape partials, and the composite were pre-specified; heterogeneity and stress-conditional cuts are exploratory. With only 7–9 crypto formation dates we lean on the well-powered equity beds (54–66) for the null that shape adds nothing, and read crypto-only positives as suggestive (App. B).

3 THE VERDICT: NOTHING BEATS VOLATILITY

Table 1 ranks each metric against volatility standalone. After BH correction, the only metrics that significantly beat volatility on *any* bed are two volatility variants (downside deviation, VaR) on the single crypto-bull bed — and they reverse on equity, where volatility significantly beats them. Every genuinely different, shape-based metric is significantly *worse* than volatility on every bed. There is no metric that consistently or generalizably beats volatility across regimes and asset classes. We fix that bar in advance to avoid a moving goalpost: an edge counts as *generalizable* only if the metric BH-significantly beats volatility on at least one crypto *and* one equity bed — i.e., survives a change of asset class. No metric clears it. This is a narrower and more defensible statement than “nothing beats volatility”: small, regime-specific edges exist for volatility’s own variants, but they do not survive a change of asset class.

4 THE ANATOMY: MAGNITUDE, NOT SHAPE

A verdict is not an explanation: if nothing beats volatility, what is left to forecast beyond it? Figure 1 plots the partial Spearman $\rho(\text{metric, drawdown} \mid \text{volatility})$ for every metric and bed, and the

picture splits cleanly by *what a metric measures*. Tail-magnitude measures carry a small but genuine increment (VaR significant on **all five beds**, partial ρ up to 0.19; downside deviation on the three crypto beds), while tail-shape measures — the Viole–Nawrocki ratio, lower-tail dependence, the tail exponent, disappointment aversion — are ≈ 0 or *negative* everywhere. This is not merely an *own-asset vs systematic* artifact: the one own-asset shape metric (the Hill tail exponent) is itself ≈ 0 /negative, so magnitude beats shape *within* the own-asset family, while no systematic co-movement metric adds signal (Figure 1 groups both axes). Nor is the increment an artifact of the -100% delisting code (mid-window delistings are $\leq 1\%$ of asset-dates; valuing them at last price or dropping them moves VaR’s increment by ≤ 0.02 ; App. H). Exact values and power checks are in App. B.

Two clarifications. An apparent paradox: VaR is *worse* than volatility standalone on equity (Table 1) yet has a positive *increment* there — as a standalone ranker it is a noisier proxy for volatility, but the partial correlation isolates its small orthogonal component. And the increment is larger where markets are fat-tailed and blow-up-prone (crypto increments 0.12–0.19, excess kurtosis 8–21, vs equity 0.03–0.08; App. F, G); tail magnitude carries incremental information precisely when the tail is active.

5 WHY VOLATILITY WINS — AND CAN ANYTHING BEAT IT?

Persistence. Volatility wins because it is *persistent*, the founding fact of ARCH/GARCH [7, 11]: trailing volatility forecasts forward volatility (rank correlation 0.42–0.84), which in turn drives forward drawdown (App. C). We resist calling forward volatility a *sufficient statistic*: a naive same-window mediation suggests it, but that test is contaminated (forward volatility and drawdown share a window). Measured on a disjoint, predictive window, trailing volatility retains a partial correlation of 0.32–0.34 on equity (App. D) — so persistence is a *leading but incomplete* explanation, not a mechanism that reduces drawdown to one number.

Does the edge pay? A rank correlation is not a profit-and-loss statement. The pre-registered composite the decomposition implies — $z(\text{vol}) + z(\text{VaR})$, equal weight, no fitting — shows *no* significant out-of-sample gain over volatility on a locked panel (Newey–West $t = 0.67$), beating it only on fat-tailed crypto-bull (App. I). A cost-aware tercile backtest agrees: the high-risk tercile realizes 10–30 percentage points more forward drawdown than the low-risk tercile on every bed, net of a 20 bps turnover cost, and the composite adds nothing net of turnover except, again, on crypto-bull. Volatility is the workhorse; magnitude adds a little only where tails are fat.

Can a machine find more? If the residual is this small, a tireless automated search should fail out-of-sample and merely overfit validation. It does, in magnitude: an LLM agent rewriting a scoring function and a 20,000-configuration sweep both inflate the best *validation* score with search intensity while the *locked-test* score plateaus at the volatility baseline. Across 12 holdout panels $\times$ 3 search seeds (App. J), the validation-selected metric’s out-of-sample edge over volatility is at most +0.07 and averages +0.02–0.03, straddling zero — no search, LLM or brute-force, finds a genuine improvement; a non-LLM random sweep reproduces

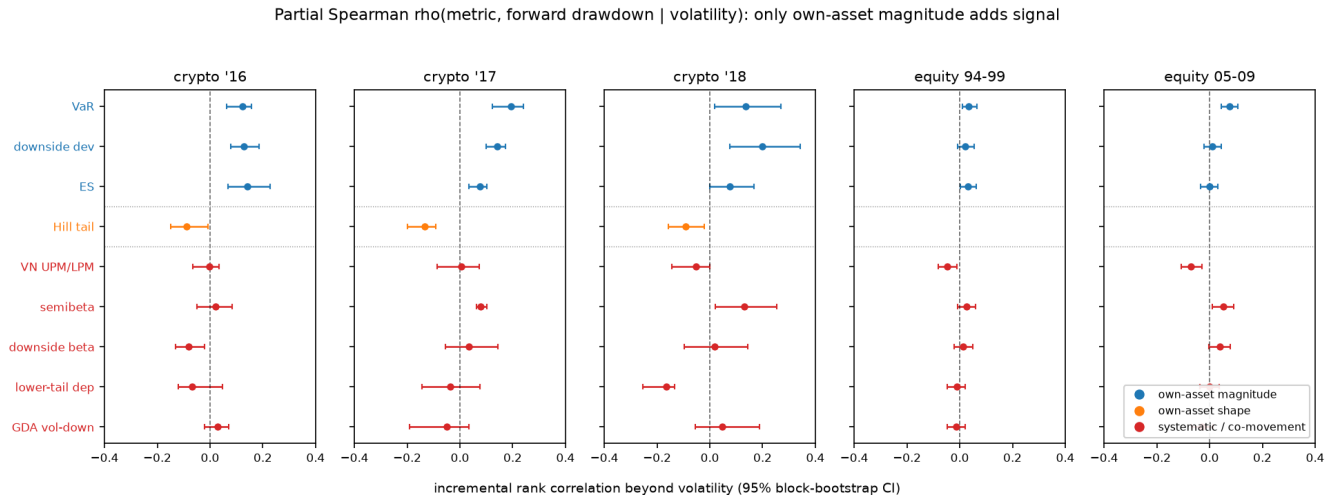


Figure 1: Incremental signal beyond volatility, by bed: partial Spearman $\rho(\text{metric, drawdown} \mid \text{volatility})$, grouped on both axes a skeptic cares about — what the metric measures (magnitude vs shape) and whose distribution it uses (own-asset vs systematic co-movement with the market). Only own-asset magnitude (blue) sits reliably right of zero. The lone own-asset shape metric, the Hill tail exponent (orange), is ≈ 0 or negative — so “magnitude beats shape” holds *within* the own-asset family — and no systematic co-movement metric (red) adds signal. Whiskers are 95% block-bootstrap CIs; dotted lines separate the three families.

the plateau, so the danger is the cheap search, not the language model. The statistics are classical data snooping [5, 21]; what is new is that an autonomous agent removes the last brake on trial count — analyst effort — so a locked, survivorship-free holdout scored exactly once becomes mandatory, not merely advisable.

What volatility is for. The clearest positive use is forecasting outright blow-ups: on the locked crypto test, volatility predicts which assets crater (forward return $\leq -80\%$) with ROC-AUC 0.68 (0.95 at a -90% cutoff) and a $2.5\times$ top-decile lift, while a gradient-boosting model over all ten metrics overfits and loses (AUC 0.59). Again the *size*, not the *shape*, of recent moves carries the signal — a ranking result, not a deployment recommendation (cost, liquidity, and reflexivity frictions are unmodeled).

6 DISCUSSION AND CONCLUSION

The pieces form one coherent picture. Volatility sets a high bar because it is persistent (§5); the only thing left to add is the magnitude of tail losses, and only a little, mostly in fat-tailed markets (§4); so a pre-registered composite yields no significant out-of-sample gain, no standalone metric beats volatility (§3), automated search cannot find one, and the economics track the statistics (§5). The recurring failure of *asymmetry* and *shape* metrics — including the partial-moment ratio that motivated this study — is not an implementation accident: once symmetric volatility is accounted for, the asymmetric structure adds no incremental cross-sectional information for forecasting drawdown. We are deliberately careful about the *strength* of the persistence story: an earlier version of this project claimed forward volatility was a sufficient statistic, found that claim was a same-window measurement artifact, and retracted it here — just as an earlier “survivorship masks downside value”

thesis was retracted when a hardened test refuted it. The paper reports only claims that survived stress-testing.

The practical takeaway is unglamorous but firm: for cross-sectional drawdown forecasting, use volatility — optionally with a small tail-magnitude term in fat-tailed markets — and treat any downside metric “discovered” to beat it, by hand or by machine, as unproven until it clears a locked, survivorship-free holdout. On a level field, volatility is not glamorous, but it is very hard to beat.

Limitations. Skill is rank-correlation, not a full tradable P&L (§5 is a first economic check, not a portfolio study); the crypto beds have few formation dates (7–9), so their significance is suggestive and the null “shape adds nothing” leans on the well-powered equity beds; the repurposed pricing metrics are daily adaptations, and results speak to drawdown forecasting, not the originals’ pricing claims.

REFERENCES

- [1] T. Andersen, T. Bollerslev, F. Diebold, and P. Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003.
- [2] A. Ang, J. Chen, and Y. Xing. Downside risk. *Review of Financial Studies*, 19(4):1191–1239, 2006.
- [3] Y. Atilgan, T. Bali, K. Demirtas, and A. Gunaydin. Left-tail momentum. *Journal of Financial Economics*, 135(3):725–753, 2020.
- [4] D. Bailey and M. López de Prado. The deflated Sharpe ratio. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- [5] D. Bailey, J. Borwein, M. López de Prado, and Q. Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–70, 2017.
- [6] V. Bawa. Optimal rules for ordering uncertain prospects. *Journal of Financial Economics*, 2(1):95–121, 1975.
- [7] T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

- [8] T. Bollerslev, A. Patton, and R. Quaedvlieg. Realized semibetas. *Journal of Financial Economics*, 144(1):227–246, 2022.
- [9] F. Chabi-Yo, S. Ruenzi, and F. Weigert. Crash sensitivity and the cross-section of expected stock returns. *Journal of Financial and Quantitative Analysis*, 53(3):1059–1100, 2018.
- [10] A. Chekhlov, S. Uryasev, and M. Zabarankin. Drawdown measure in portfolio optimization. *International Journal of Theoretical and Applied Finance*, 8(1):13–58, 2005.
- [11] R. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of U.K. inflation. *Econometrica*, 50(4):987–1007, 1982.
- [12] A. Farago and R. Tédongap. Downside risks and the cross-section of asset returns. *Journal of Financial Economics*, 129(1):69–86, 2018.
- [13] P. Fishburn. Mean-risk analysis with risk associated with below-target returns. *American Economic Review*, 67(2):116–126, 1977.
- [14] C. Harvey, Y. Liu, and H. Zhu. . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.
- [15] B. Kelly and H. Jiang. Tail risk and asset prices. *Review of Financial Studies*, 27(10):2841–2871, 2014.
- [16] Y. Levi and I. Welch. Symmetric and asymmetric market betas and downside risk. *Review of Financial Studies*, 33(6):2772–2795, 2020.
- [17] M. Magdon-Ismael, A. Atiya, A. Pratap, and Y. Abu-Mostafa. On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1):147–161, 2004.
- [18] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.
- [19] F. Sortino and R. van der Meer. Downside risk. *Journal of Portfolio Management*, 17(4):27–31, 1991.
- [20] F. Viole and D. Nawrocki. Predicting risk/return performance using upper partial moment/lower partial moment metrics. *Journal of Mathematical Finance*, 6(5):900–920, 2016.
- [21] H. White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000.

A REPRODUCIBILITY

Code, data pointers, and result logs to reproduce every table and figure are available at <https://github.com/meghanadhpulivarthi/downside-risk-metrics-study>. Data sources: a CoinMarketCap daily crypto dump (retaining delisted coins), yfinance S&P 500 constituents and the market index, and point-in-time index membership.

B PARTIAL CORRELATIONS AND POWER

Table 2 gives the exact partial correlations plotted in Figure 1, with BH significance. VaR survives on all five beds; the one equity ES cell that is nominally significant does not survive correction. Power on the thin crypto beds: for VaR, on crypto-16 ($n = 8$) and crypto-17 ($n = 9$) the effect (0.12, 0.19) exceeds the 80%-power minimum detectable effect (0.09, 0.09) and is stable under leave-one-date-out; on crypto-18 ($n = 7$) the effect (0.13) is below its MDE (0.18) and is reported as suggestive. A placebo (permuting forward drawdown within each date) drives the trailing-volatility correlation to ≈ 0 on four of five beds; the exception, crypto-18 (+0.028, CI [0.003, 0.059]), is negligible against a real skill of ≈ 0.38 and reflects the coarse block bootstrap at $n = 7$.

C THE PERSISTENCE CHAIN

Table 3 decomposes why trailing volatility forecasts drawdown. Both links exceed the direct trailing-vol→drawdown skill they

Table 2: Partial Spearman $\rho(\text{metric, drawdown} \mid \text{volatility})$. Rows 1–3 tail magnitude, rows 4–9 tail shape. * survives BH ($q = 0.05$, positive); † survives BH negative; n/e = not estimable per date.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
VaR	+0.12*	+0.19*	+0.13*	+0.03*	+0.08*
downside dev.	+0.13*	+0.14*	+0.20*	+0.02	+0.01
ES [3]	+0.14*	+0.08*	+0.08	+0.03	+0.00
neg. semibeta [8]	+0.02	+0.08*	+0.13*	+0.03	+0.05*
UPM/LPM [20]	−0.00	+0.01	−0.05	−0.05†	−0.07†
downside beta [2]	−0.08†	+0.03	+0.02	+0.01	+0.04
Hill tail [15]	−0.09†	−0.13†	−0.09†	n/e	n/e
lower-tail dep. [9]	−0.07	−0.04	−0.17†	−0.01	+0.00
GDA [12]	+0.03	−0.05	+0.05	−0.01	−0.03

Table 3: Persistence chain (mean per-date Spearman; all CIs exclude 0).

Bed	trail→fwd vol	fwd vol→dd	(trail→dd)
crypto-16	0.68	0.77	0.62
crypto-17	0.42	0.62	0.36
crypto-18	0.42	0.53	0.38
equity 94	0.84	0.71	0.61
equity 05	0.80	0.70	0.56

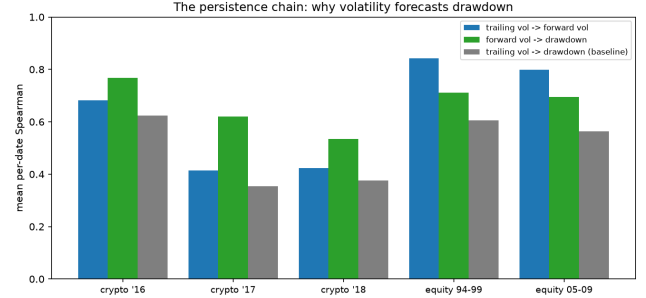


Figure 2: The two links of the persistence chain relative to the direct trailing-volatility→drawdown skill, by bed.

compose. The forward-vol→drawdown column is contemporaneous (both over the same forward window) and hence partly mechanical [1]; the genuinely predictive link is trailing→forward volatility (see §5 and Figure 2).

D MEDIATION: FORWARD VOLATILITY IS NOT A SUFFICIENT STATISTIC

It is tempting to claim drawdown predictability simply is volatility persistence — that forward volatility is a sufficient statistic. A naive mediation test seems to confirm it: controlling for forward volatility, trailing volatility’s partial correlation with drawdown collapses to ≈ 0 on equity. But that test is contaminated: forward

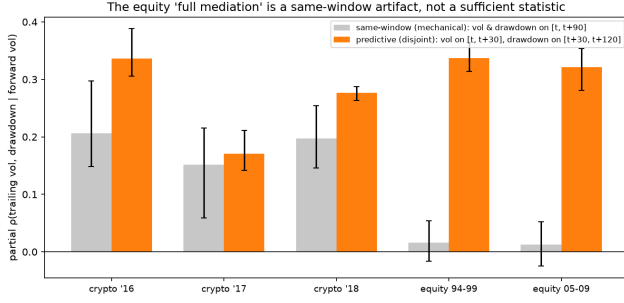


Figure 3: Mediation: partial $\rho(\text{trailing vol, drawdown} \mid \text{forward vol})$. Under the same-window (mechanical) design it is ≈ 0 on equity, falsely suggesting full mediation; under the predictive (disjoint-window) design it is 0.32–0.34 — trailing volatility is not screened off, so forward volatility is not a sufficient statistic.

volatility and forward drawdown are measured over the *same* window, so conditioning on one mechanically absorbs the other. Measuring the mediator on a *disjoint, earlier* window — volatility over days 1–30, drawdown over days 31–120, so it is genuinely predictive — the mediation vanishes: trailing volatility retains a partial correlation of 0.32–0.34 on equity (Figure 3). Trailing volatility carries forecasting content a short-horizon volatility forecast does not, so persistence is a leading but incomplete explanation. Consistently, drawdown skill and volatility persistence rise and fall together as the horizon varies (App. E).

E HORIZON DYNAMICS

Sweeping the forecast horizon from 14 to 90 days, volatility persistence and drawdown skill move together (Figure 4): on equity both rise (persistence 0.66 $\rightarrow$ 0.84, skill 0.47 $\rightarrow$ 0.61, at a roughly constant gap); on crypto both are flat. Wherever persistence is higher, so is skill — consistent with persistence driving most of the forecastable variation, with the constant gap the part it does not explain.

F CROSS-BED HETEROGENEITY

Figure 5 relates the VaR increment to each bed’s tail fatness. The increment is systematically larger in the fat-tailed, blow-up-prone crypto beds than in equity; with only five beds this is descriptive, but the direction is consistent with the conditional evidence in Appendix G.

G CONDITIONAL INCREMENT (STRESS VS. CALM)

Pooling dates by asset class and splitting each pool at its median aggregate (cross-sectional-median) volatility, the VaR increment concentrates in high-stress dates within crypto — rising from +0.13 (CI [0.06, 0.18]) in low-stress dates to +0.18 ([0.09, 0.24]) in high-stress dates (24 dates, 12 per regime; suggestive given the overlap) — while on equity it is flat and small (+0.056 vs. +0.050; 120 dates). Tail magnitude carries incremental information precisely when the tail is active.

H DELISTING-CODING ROBUSTNESS

The crypto forward-drawdown codes a mid-window delisting as a terminal -100% loss. To confirm the magnitude increment is genuine crash-severity skill — not an artifact of that code, nor a restatement of the blow-up result (§5) — we recompute $\rho(\text{metric, drawdown} \mid \text{vol})$ under two alternative codings: valuing a delisted asset at its *last traded price* (ordinary peak-to-trough, no -100% override), and *excluding* delisted assets (skill among survivors only). Within the point-in-time top-100 universe, mid-window delistings are rare — 0.9% of asset-dates on crypto-2016 and none on the other four beds — so the three codings barely differ. VaR’s increment is unchanged to within ± 0.02 across all three (Table 4); downside deviation and ES behave identically. The magnitude edge is drawdown-*severity* forecasting, distinct from the terminal-death signal.

Table 4: VaR partial $\rho(\text{VaR, drawdown} \mid \text{vol})$ under three delisting codings. Baseline and last-price coincide (mid-window delistings are rare); the increment is unchanged to within ± 0.02 .

Bed	baseline (-100%)	last-price	exclude
crypto-16	+0.12	+0.12	+0.14
crypto-17	+0.19	+0.19	+0.19
crypto-18	+0.14	+0.14	+0.14
equity-94	+0.035	+0.035	+0.035
equity-05	+0.076	+0.076	+0.076

I PRE-REGISTERED COMPOSITE AND ECONOMIC BACKTEST

The composite is $z(\text{vol}) + z(\text{VaR})$, equal weight, computed cross-sectionally per date, with no fitting. VaR was chosen a priori as the magnitude metric surviving correction on all five beds (§4); the weights are untuned, so the fully out-of-sample evidence is the locked-panel test (Table 5), which is not significant. The economic backtest (§5, Figure 6) sorts the universe into risk terciles and measures realized-drawdown discrimination (forward DD of the high- minus low-risk tercile) and the forward-return spread, net of a 20 bps turnover cost.

J AUTOMATED DISCOVERY: BEST-OF- N AND ROBUSTNESS

On the original locked panel, both the LLM loop and the 20,000-config sweep inflate the best *validation* score with search intensity while the locked *test* score stays flat at the volatility baseline (Table 6, Figure ??): validation-selected winners beat volatility on the locked test only 43% of the time (directed search) despite a 0.94 validation–test rank correlation.

To check that this is not a single-panel artifact, we re-ran the candidate search (a general nonlinear generator over per-date feature ranks, 1,500 candidates per run) over a grid of holdout panels $\times 3$ search seeds, in two families: *out-of-time* — a fixed-length test window slid through the timeline with validation always chronologically earlier (six windows, including the original 2017 $\rightarrow$ 2018

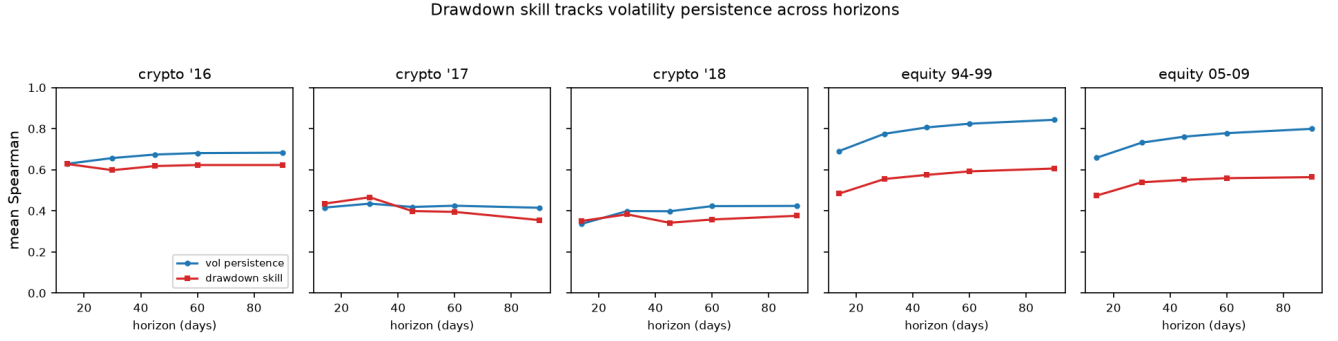


Figure 4: Volatility persistence (blue) and drawdown-forecast skill (red) versus forecast horizon, by bed. The curves track.

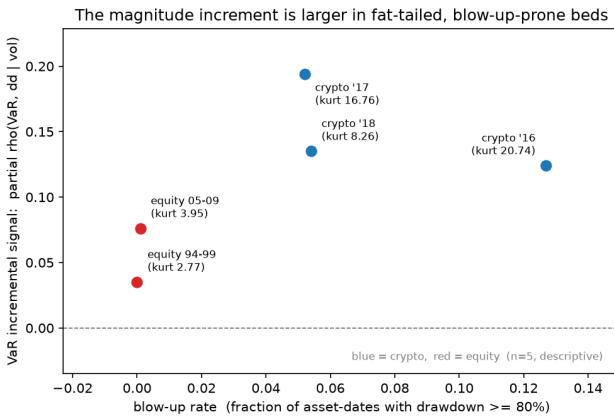


Figure 5: VaR incremental signal (partial ρ | volatility) versus blow-up rate, by bed. Larger in fat-tailed crypto; descriptive ($n = 5$).

Table 5: Composite $z(\text{vol}) + z(\text{VaR})$ vs. volatility: per-date Spearman and difference (95% CI). Significant only on crypto-bull; locked-panel test not significant.

Test bed	vol ρ	comp ρ	difference (95% CI)
crypto-16	0.62	0.62	-0.00 (n.s.)
crypto-17	0.36	0.40	+0.05 [0.02, 0.07]
crypto-18	0.38	0.39	+0.01 (n.s.)
equity 94	0.61	0.60	-0.008 [-0.011, -0.004]
equity 05	0.56	0.56	-0.00 (n.s.)
Locked panel: 0.376 $\rightarrow$ 0.388, Newey-West $t = 0.67$ (n.s.).			

split) — and *random* — order-ignoring 10-val/7-test repartitions of the non-train dates (six draws). Per run we record the validation-selected metric's locked-test edge over volatility (*test gap*) and the fraction of validation-winners that also beat volatility on test (*win-rate*).

The plateau is stable. The test gap averages +0.020 (out-of-time; range -0.028 to +0.046) and +0.035 (random; +0.014 to +0.072) — always small and often straddling zero, so no search beats

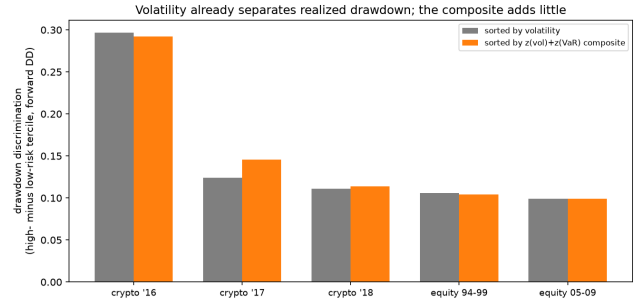


Figure 6: Realized drawdown discrimination (high- minus low-risk tercile) by bed, sorting on volatility vs. the composite. Volatility already separates drawdown strongly; the composite is indistinguishable except marginally on crypto-bull.

Table 6: Best-of- N discovery. Validation inflates with search intensity; the locked test does not (volatility baseline 0.376). Validation and test are different splits, so compare trends, not levels.

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373

volatility out-of-sample by more than noise on any panel or seed. Validation-test rank correlation is 0.94–0.99 throughout. The win-rate is not. It averages 0.79 (out-of-time, range 0.55–1.00) and 0.82 (random, 0.62–0.97), lowest on holdouts where volatility's own skill is high on the test (the 2018 crash: 0.43 directed / 0.57 milder generator) and near 1.0 on calm windows. Because these nominal “wins” come with a test gap of essentially zero, the sign of the win-rate is far less informative than its magnitude: the validation-selected metric matches, but does not beat, volatility.

